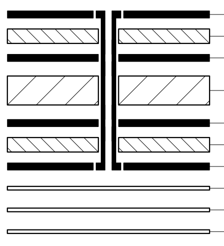
Main board Fabrication Document

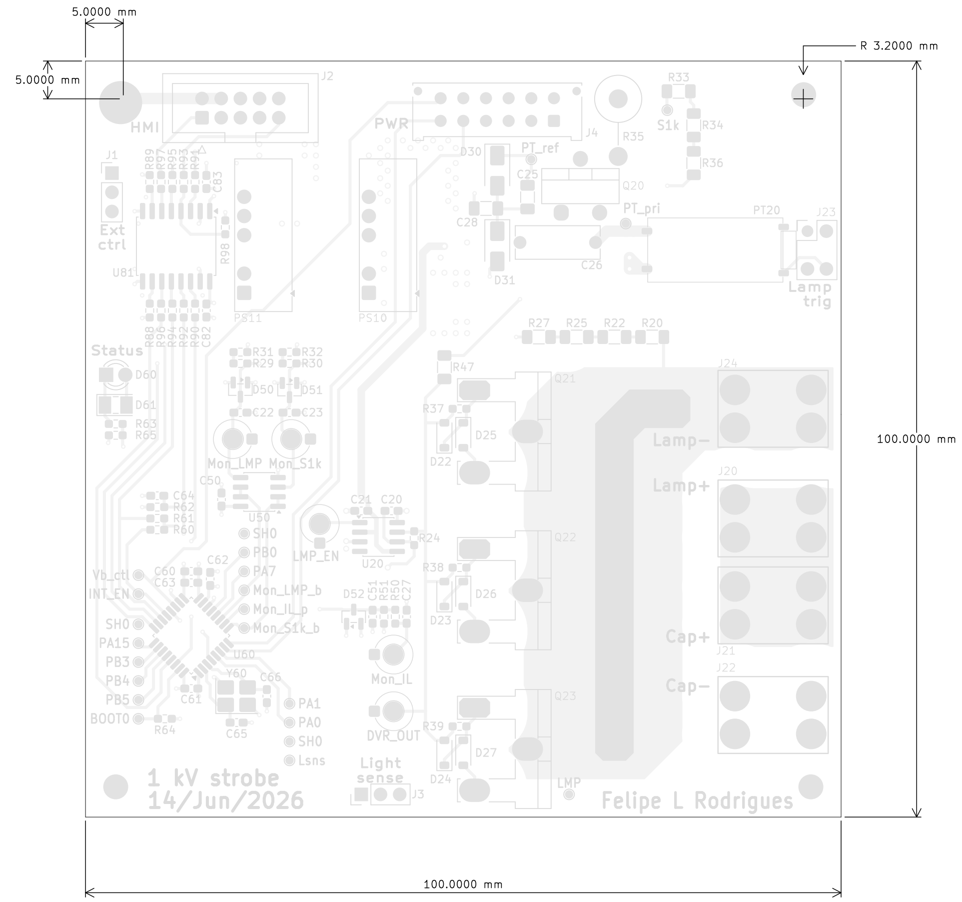
Top Fabrication (Scale 1:1)

Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**F.Cu (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**In1.Cu (GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Core		1.828mm	FR4	Dielectric	
Copper	**In2.Cu (Sig, PWR)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**B.Cu (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.
2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
4) SURFACE FINISH: HAL SNPB
5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
9) PCB MATERIAL REQUIREMENTS:
   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

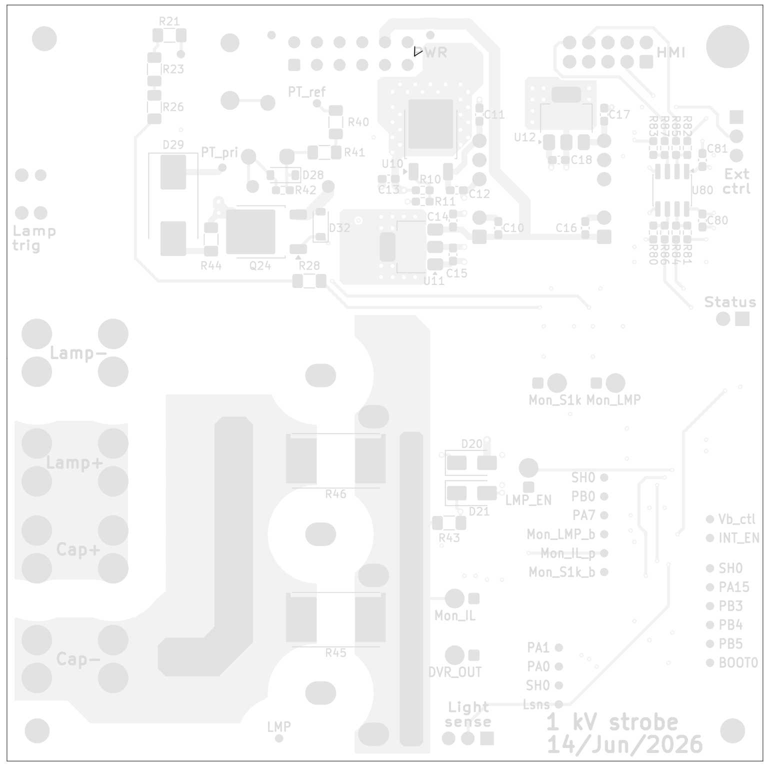
BOARD SIZE	100.000 × 100.000 mm
BOARD THICKNESS	2.278 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	1.000 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.300 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 16e6ec6
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 1 of 10

Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



1 kV strobe
04/Jul/25
Felipe L Rodrigues

Comments:	Company: ChurrasCod Electronics	Variant: PRELIMINARY	Git Hash: 16e6ec6
	Board Name: **Main board**	Project Name: **1 kV strobe**	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **2** of **10**

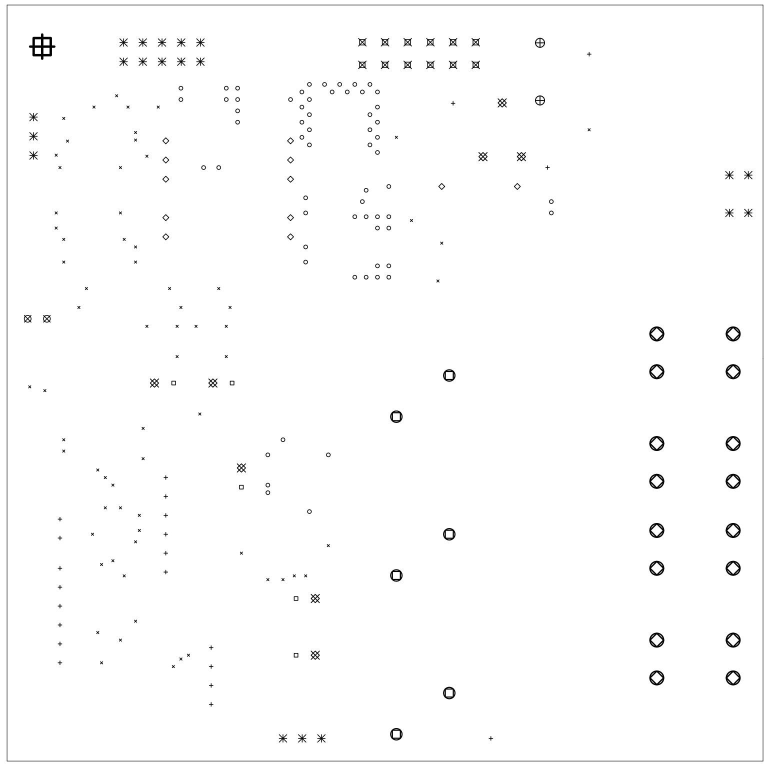
ChurrasCod Electronics

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	69	0.30mm (11.81mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Via
○	62	0.50mm (19.69mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Via
+	22	0.50mm (19.69mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
□	5	0.70mm (27.56mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
◇	12	0.80mm (31.50mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⨂	2	0.90mm (35.43mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
✳	20	1.00mm (39.37mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊠	12	1.02mm (40.16mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⋈	8	1.10mm (43.31mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊕	2	1.20mm (47.24mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⧈	6	1.50mm (59.06mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
◈	16	1.80mm (70.87mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
⊞	1	3.20mm (125.98mils)	PTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Pad
	Total 237					



ChurrasCod Electronics

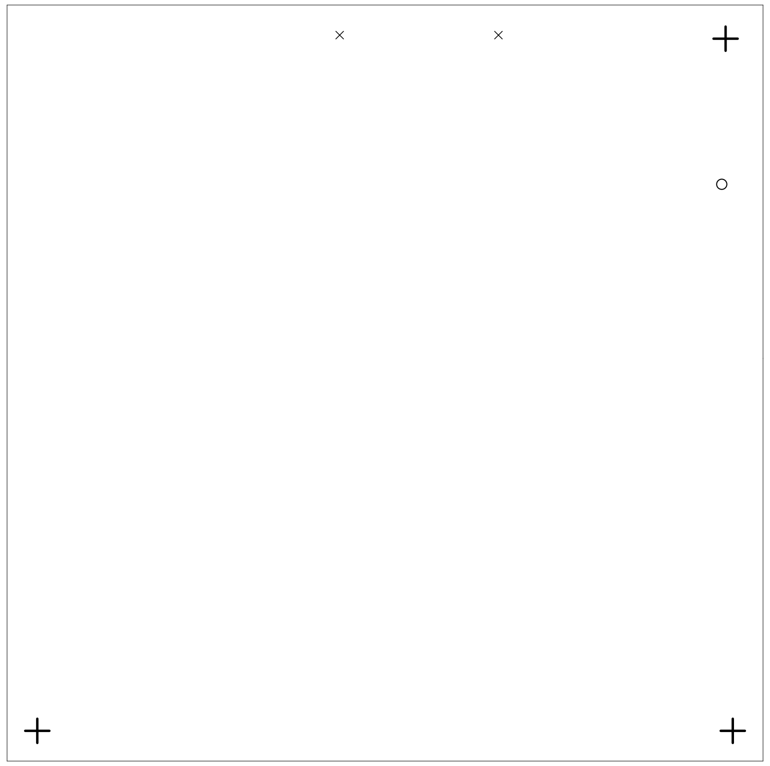
Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY		Git Hash: 16e6ec6
	Board Name: **Main board**		Project Name: **1 kV strobe**		
Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)	
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **3** of **10**	

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1.02mm (40.16mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
○	1	1.35mm (53.15mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
+	3	3.20mm (125.98mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
	Total 6					

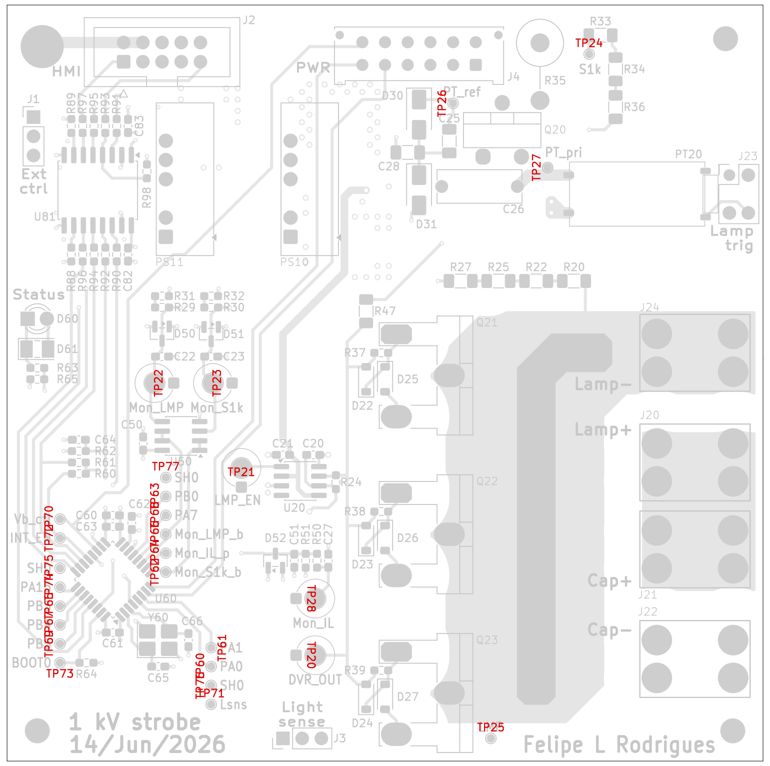


ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 16e6ec6
	Board Name: **Main board**		Project Name: **1 kV strobe**	
Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **4** of **10**

Main board Fabrication Document

Top Test Points (Scale 1:1)

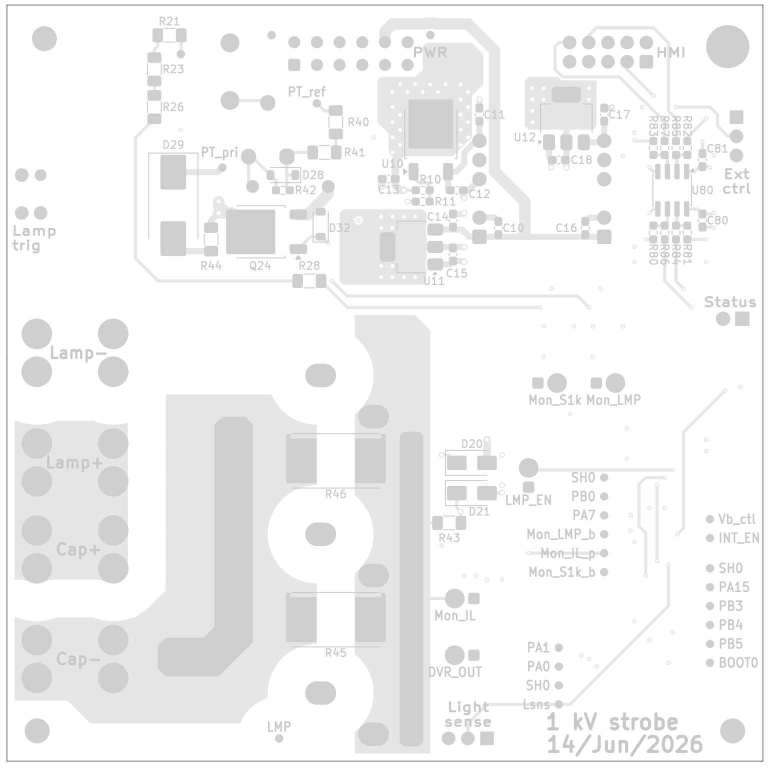


Ref.	Net	X [mm]	Y [mm]
TP20	SH0	44.50	19.00
TP21	SH0	36.00	42.50
TP22	Mon_LMP	25.77	55.00
TP23	Mon_S1k	33.50	55.00
TP24	S1k	82.00	98.50
TP25	LMP	69.00	8.00
TP26	TP_REF	64.00	92.00
TP27	TP_PRI	76.50	83.50
TP28	Mon_IL	44.50	26.50
TP60	Net-(U60-PA0)	32.00	17.50
TP61	Net-(U60-PA1)	32.00	20.00
TP62	Mon_S1k_buf	26.00	30.00
TP63	Net-(U60-PB0)	26.00	40.00
TP64	Mon_IL_prot	26.00	32.50
TP65	Net-(U60-PB3)	12.00	25.50
TP66	Mon_LMP_buf	26.00	35.00
TP67	Net-(U60-PB4)	12.00	23.00
TP68	Net-(U60-PA7)	26.00	37.50
TP69	Net-(U60-PB5)	12.00	20.50
TP70	Vbus_ctrl	12.00	37.00
TP71	Light_sns	32.00	12.50
TP72	INT_EN	12.00	34.50
TP73	BOOT0	12.00	18.00
TP74	Net-(U60-PA15)	12.00	28.00
TP75	SH0	12.00	30.50
TP76	SH0	32.00	15.00
TP77	SH0	26.00	42.50

ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics	Variant: PRELIMINARY	Git Hash: 16e6ec6
		Board Name: **Main board**	Project Name: **1 kV strobe**	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13 / Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A3 / Sheet: 5 of 10

Main board Fabrication Document

Bottom Test Points (Scale 1:1)



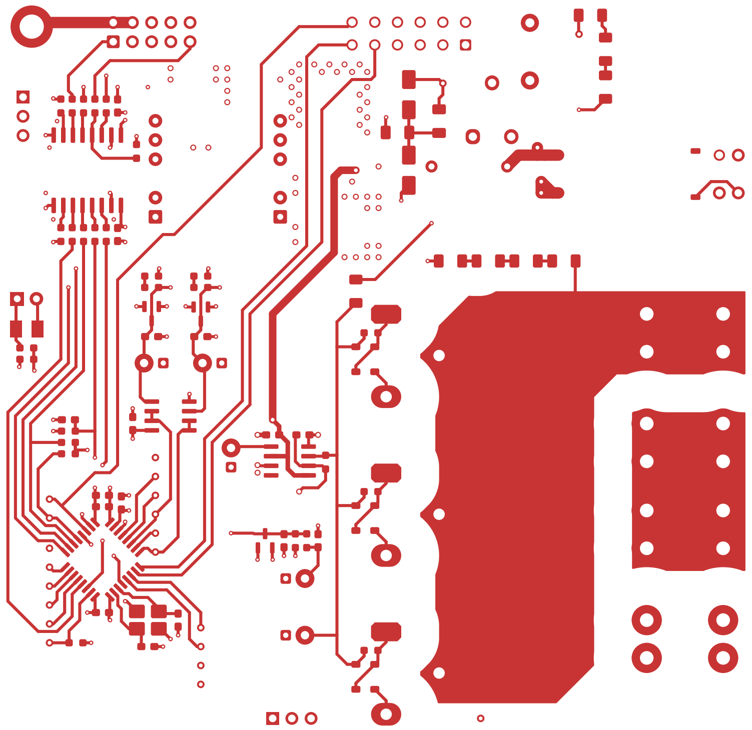
Ref.	Net	X [mm]	Y [mm]

1 kV strobe
04/Jul/25
Felipe L Rodrigues

ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics	Variant: PRELIMINARY	Git Hash: 16e6ec6
		Board Name: **Main board**	Project Name: **1 kV strobe**	
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb / Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:	Reviewer:	Size: **A3**	Sheet: **6** of **10**

Main board Fabrication Document

F.Cu (Sig, PWR) (Scale 1:1)

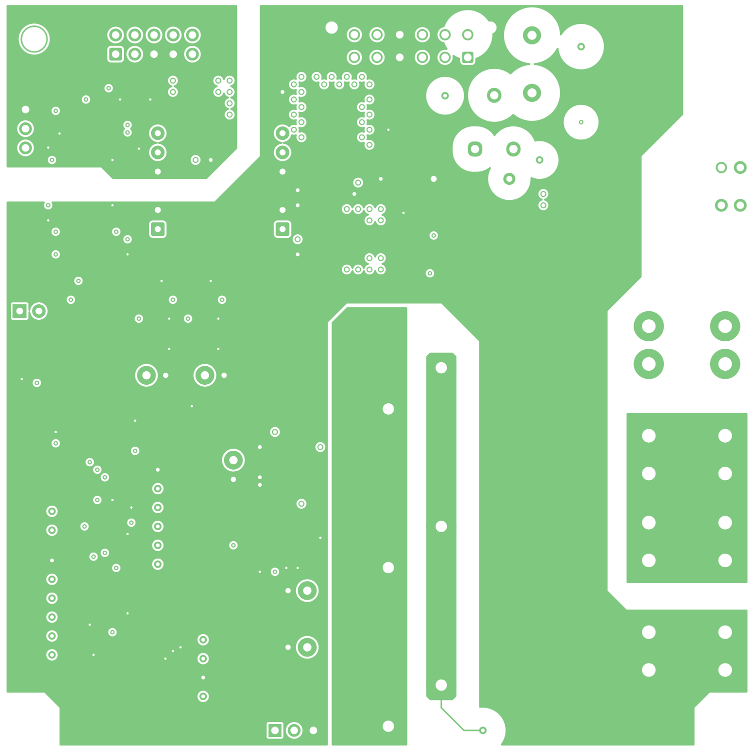


ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 16e6ec6
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: F.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 7 of 10

Main board Fabrication Document

In1.Cu (GND) (Scale 1:1)

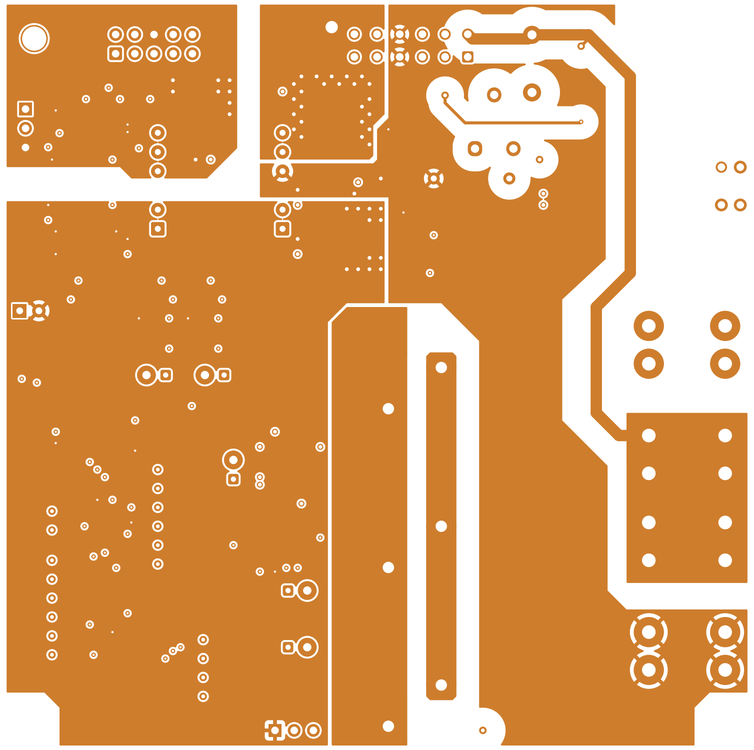


ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 16e6ec6
	Board Name: **Main board**		Project Name: **1 kV strobe**	
Sheet Title: In1.Cu (GND) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **8** of **10**

Main board Fabrication Document

In2.Cu (Sig, PWR) (Scale 1:1)

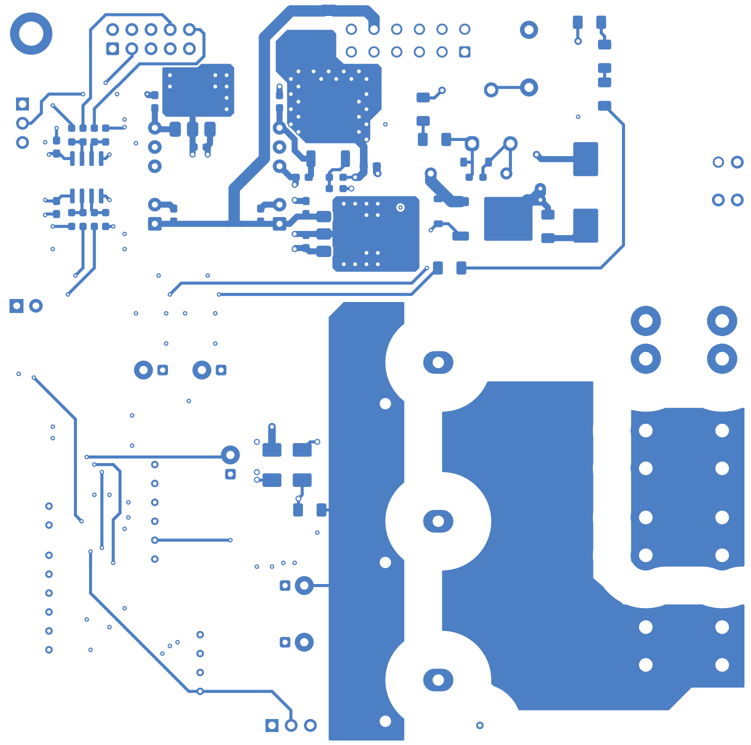


ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 16e6ec6
	Board Name: **Main board**		Project Name: **1 kV strobe**	
Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 9 of 10

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



ChurrasCod Electronics

Comments:	
Company:	ChurrasCod Electronics
Variant:	PRELIMINARY
Git Hash:	16e6ec6
Board Name:	Main board
Project Name:	1 kV strobe
Sheet Title:	B.Cu (Sig, PWR) (Scale 1:1)
File Name:	Strobe_main.kicad_pcb
Designer:	FR
Date:	2024-04-13
Revision:	1.0.0+ (Unreleased)
Sheet Path:	
Reviewer:	
Size:	A3
Sheet:	10 of 10